Capstone Weekly Progress Report 2
ChalkStream (Group #62)
[Week of Nov 7, 2025]
Supervisor: Safaa Bedawi
Group Members: Cameron Bodkin-Dick, Eric Cui, Sandy Alzabadani

1. Overview of Current Week
Finalized

2. Milestone Progress Summary

Component	Description	Current Status % Complete	Remarks / Issues
3.1 Frame Preprocessing	Isolating the board from the rest of the wall, rectifying any angular distortion, and handling dynamic obstructions (instructor walking across the board)	50%	
3.2 Content Segmentation	Determine candidate areas of written content, and assign text, equation, and diagram affinity	85%	Next steps are integrating with frame preprocessing to eliminate the simple stub functions.
3.3. Handwriting OCR	Setting up the testing & training environment for the OCR and prepping datasets.	10%	The environment for training the OCR is mostly set up to start.

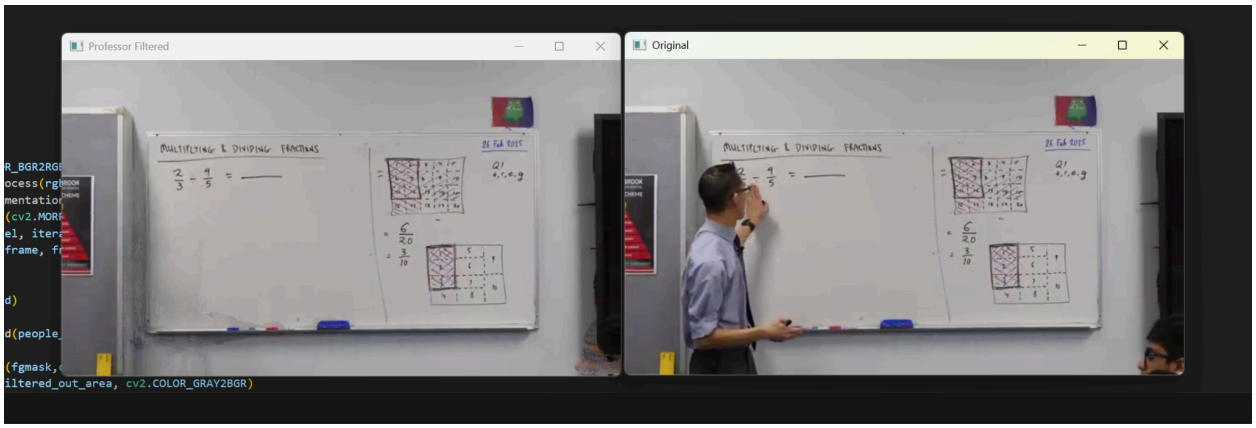
3. Individual Member Contributions
Cameron Bodkin-Dick

This Week: I refactored the Segmenter.py source file to align better with the needs for live service integration. The module now includes functionality to process an input image and output the detected and categorized regions to a specified output path, which will later be used by the Text, Equation, and Diagram Recognition modules.

Additionally, I enhanced the module's runtime behavior for continuous board capture. On subsequent calls to segment_frame() during the same session, the segmenter now compares the newly detected bounding boxes against those stored from previous frames. This comparison identifies regions that are new or have changed beyond a configurable epsilon threshold, allowing finer control over when downstream recognition models need to be re-run.

Next Week: Going to be focused on working with Eric to develop the board zoom isolation functionality and integrate with the segmenter module. Also going to try to utilize Eric's laptop's Nvidia graphics card for the GPU performance tests. Also going to be working with Sandy to help with making use of the LectureVideoDB and LectureMath datasets for training.

Eric Cui



This Week:
Updated the noise object removal script such that the filtered area uses the content from the previous frame, such that noise will be completely filtered out. Some noise may linger, but overall the filter is clear enough for capture.

Next Week:
Develop board zoom functionality and integrate the three-frame preprocessing script (Board zoom, noise filter, and greyscale). Additionally, assist Cameron in the performance test on LectureVideoDB and LectureMath.

Sandy Alzabadani

This week: Set up the environment for the handwritten OCR training. Had the python script ready for reading the dataset and extraction.

Next Week: Prototyping the handwritten OCR, examining the output accordingly and tuning it for better extraction. Performing tests to improve the reading and extraction process for a better output. Working with Cameron on LectureVideoDB and LectureMath datasets for training.